Even / odd & 1

7 → 0111
& 0001
0001 = 1 × 0
odd

Swap Two Numbers:

a = 5 0101
b = 3 0011
0110

(i) a = a ^ b = 6
(ii) b = a ^ b = 5
(iii) a = a ^ b = 3

0110
0011
0101

0110
0101
0011

n = 8, 6, 4, 3, 5

8 1000
2^3 0111
0000

if ( )
n & (n-1) == 0

6 0110
× 0101
0100

5 0101
×

5	4	3	2	1	0
2^5	2^4	2^3	2^2	2^1	2^0
32	16	8	4	2	1
1	0	0	0	0	0
		1	0	0	0
		1	0	0	0

4 0100
2^2 0011
0000

0011
0010
3 0010
×

(Switch OFF RSB) :→ Rightmost Set Bit

12 → 1100 ⇒ 1000
Turn this OFF

12
1100
3^2 2^1 2^0
8 4 2 1

n & (n-1)
12 → 1100

1100
1011
& 1000

Bit wise not (-12)
2's Complement

1100
0011
0001
0100

1100
& 0100
0100 = 4

NOT

0	1
1	0

0111
1

7 & -7

7 0111
-7 1001
0001 = 1

0111
1000 1c
0001 +1 } 2c
1001

T-F S = 0101
F-T ~S = 1010 =
(~S) = 10

$\boxed{-n-1}$
-(-1) = -6
-500 -1
-(-500) -1
⇒ 500 -1
= 499

* Check if the kth bit is set:

1st bit
(k = 1)
L ← R

0001<<1
0010

n = 10 = 1010
3 2 1 0

(n & (1<<k)) != 0

1010 & (1<<1)
1010 & 0010

bin(1) = 0001

1010
0010
0010

0100
n & (1<<k)
0100 & 0010
0100
0010
0

Set the kth Bit:

n = 10 = 1010
k = 0
3 2 1 0

o/p 1011 = 11

n | (1<<k)

1<<0 = 1

1010
0001
or 1011 = 11

Bit Masking Cheat Code :→

* n&1 — odd / even
* n&(n-1) — remove rightmost set bit
* n & -n — get " " " "
* n << 1 — multiply by 2
* n >> 1 — divide by 2
* n | (1<<k) — set k^{th} bit
* n & ~(1<<k) — clear k^{th} bit
* n ^ (1<<k) — toggle k^{th} bit
* n & (1<<k) — check if k^{th} bit is set